

Paragraph Proofs and Flowchart Proofs

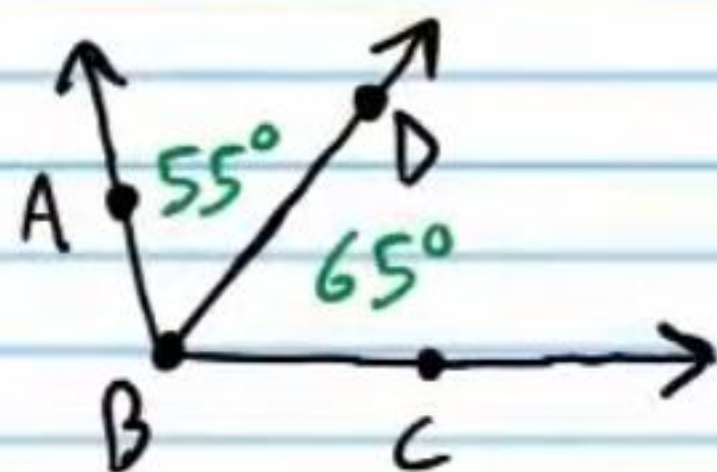
Name
Date
Course

Paragraph Proofs

Paragraph Proof: A proof demonstrated in written form.

Convert the following proofs from two-column form to Paragraph form.

① If $m\angle ABD = 55^\circ$ and $m\angle DBC = 65^\circ$, then $m\angle ABC = 120^\circ$.



Two-Column Form

Statement	Reason
I) $m\angle ABD = 55^\circ$ & $m\angle DBC = 65^\circ$	Given.
II) $m\angle ABC = m\angle ABD + m\angle DBC$	$\angle$ Add Post.
III) $m\angle ABC = 55^\circ + 65^\circ$	Subs. Prop of =.
IV) $m\angle ABC = 120^\circ$	Simplification.

Paragraph Form

It is given that $m\angle ABD = 55^\circ$ and $m\angle DBC = 65^\circ$ and by the Angle Addition Postulate, $m\angle ABC = m\angle ABD + m\angle DBC$. Also, by the Substitution Property of Equality, $m\angle ABC = 55^\circ + 65^\circ$. This equation simplifies to $m\angle ABC = 120^\circ$.

② If $AB = 15$ and $BC = 3$ in the segment below, then $AC = 18$.



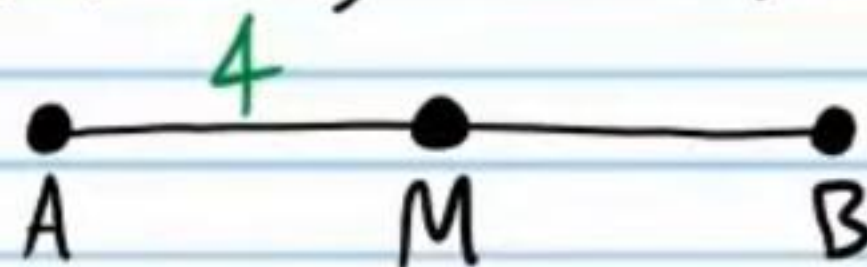
Two-Column Form

Statement	Reason
I) $AB = 15$ & $BC = 3$	Given.
II) $AC = AB + BC$	Seg Add. Post.
III) $AC = 15 + 3$	Subs. Prop of =.
IV) $AC = 18$	Simplification.

Paragraph Form

It is given that $AB = 15$ and $BC = 3$ and by the Segment Addition Postulate, $AC = AB + BC$. $AC = 15 + 3$ by the Substitution Property of Equality and this equation simplifies to $AC = 18$.

③ If M is the midpoint of the segment below, and $AM = 4$, then $AB = 8$.



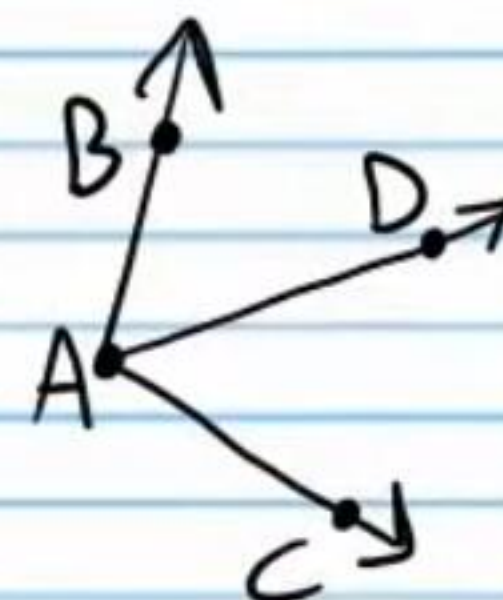
Two-Column Form

Statement	Reason
I) M is the midpoint of $\overline{AB}$.	Given.
II) $AM = MB$	Def. of midpoint.
III) $AM = 4$	Given.
IV) $MB = 4$	Trans. Prop of =.
V) $AB = AM + MB$	Seg Add Post.
VI) $AB = 4 + 4$	Subs. Prop of =.
VII) $AB = 8$	Simplification.

Paragraph Form

It is given that M is the midpoint of $\overline{AB}$ and by the definition of the midpoint, $AM = MB$. It is also given that $AM = 4$ and so by the Transitive Property of Equality $MB = 4$. By the Segment Addition Property, $AB = AM + MB$, and therefore by the Substitution Property of Equality, $AB = 4 + 4$ and this equation simplifies to $AB = 8$.

④ If $\overrightarrow{AD}$ bisects $\angle BAC$, then $m\angle BAC = 2m\angle CAD$.



Two-Column Form

Statement	Reason
I) $\overrightarrow{AD}$ bisects $\angle BAC$.	Given.
II) $m\angle BAD = m\angle CAD$	Def. of $\angle$ bisector.
III) $m\angle BAC = m\angle BAD + m\angle CAD$	$\angle$ Add Post.
IV) $m\angle BAC = m\angle CAD + m\angle CAD$	Subs. Prop of =.
V) $m\angle BAC = 2m\angle CAD$	Simplification.

Paragraph Form

It is given that $\overrightarrow{AD}$ bisects $\angle BAC$ and so by the definition of an angle bisector, $m\angle BAD = m\angle CAD$. The Angle Addition Postulate states that, $m\angle BAC = m\angle BAD + m\angle CAD$ and by the Substitution Property of Equality, $m\angle BAC = m\angle CAD + m\angle CAD$. Finally, this equation simplifies to $m\angle BAC = 2m\angle CAD$.

Write paragraph proofs for the following statements

⑤ If $\angle A$ and $\angle B$ are both straight angles, then they are congruent.

It is given that $\angle A$ and $\angle B$ are both straight angles, so $m\angle A = 180^\circ$ and $m\angle B = 180^\circ$ by the definition of straight angles. Consequently, $m\angle A = m\angle B$ by the Transitive Property of Equality. Therefore, $\angle A \cong \angle B$ by the definition of congruent angles.

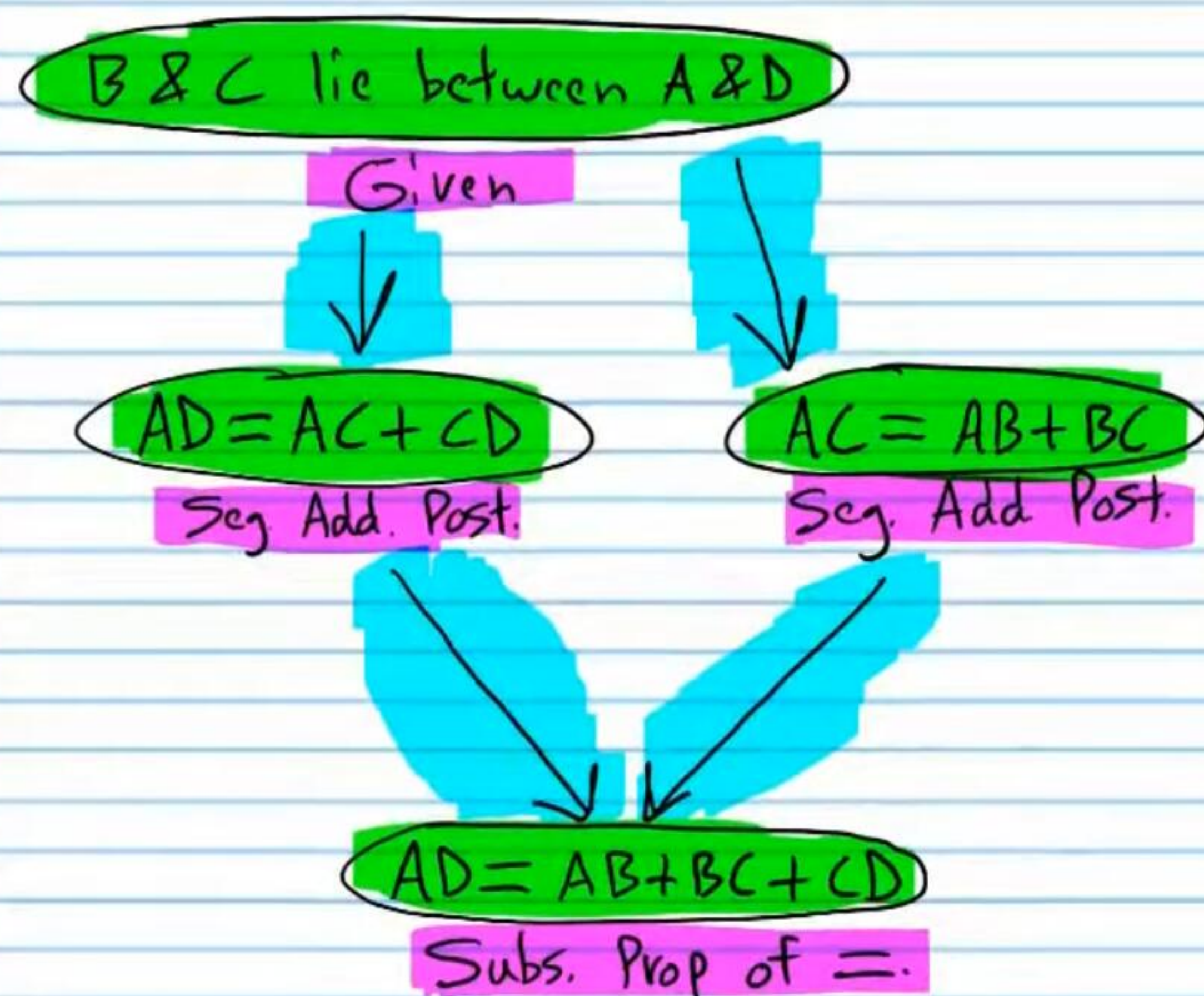
⑥ If $\angle A$ and $\angle B$ are both right angles, then they are congruent.

It is given that $\angle A$ and $\angle B$ are both right angles and so by the definition of right angles, $m\angle A = 90^\circ$ and $m\angle B = 90^\circ$. By the Transitive Property of Equality, $m\angle A = m\angle B$ and by the definition of congruent angles, $\angle A \cong \angle B$.

Flowchart Proofs

Flowchart Proof: A proof that uses **arrows** to show flow of thoughts, **bubbles** to show statements, and **spaces** under the bubbles to show reasons for the statements.

Example



Convert the following proofs from two-column form to flowchart form.

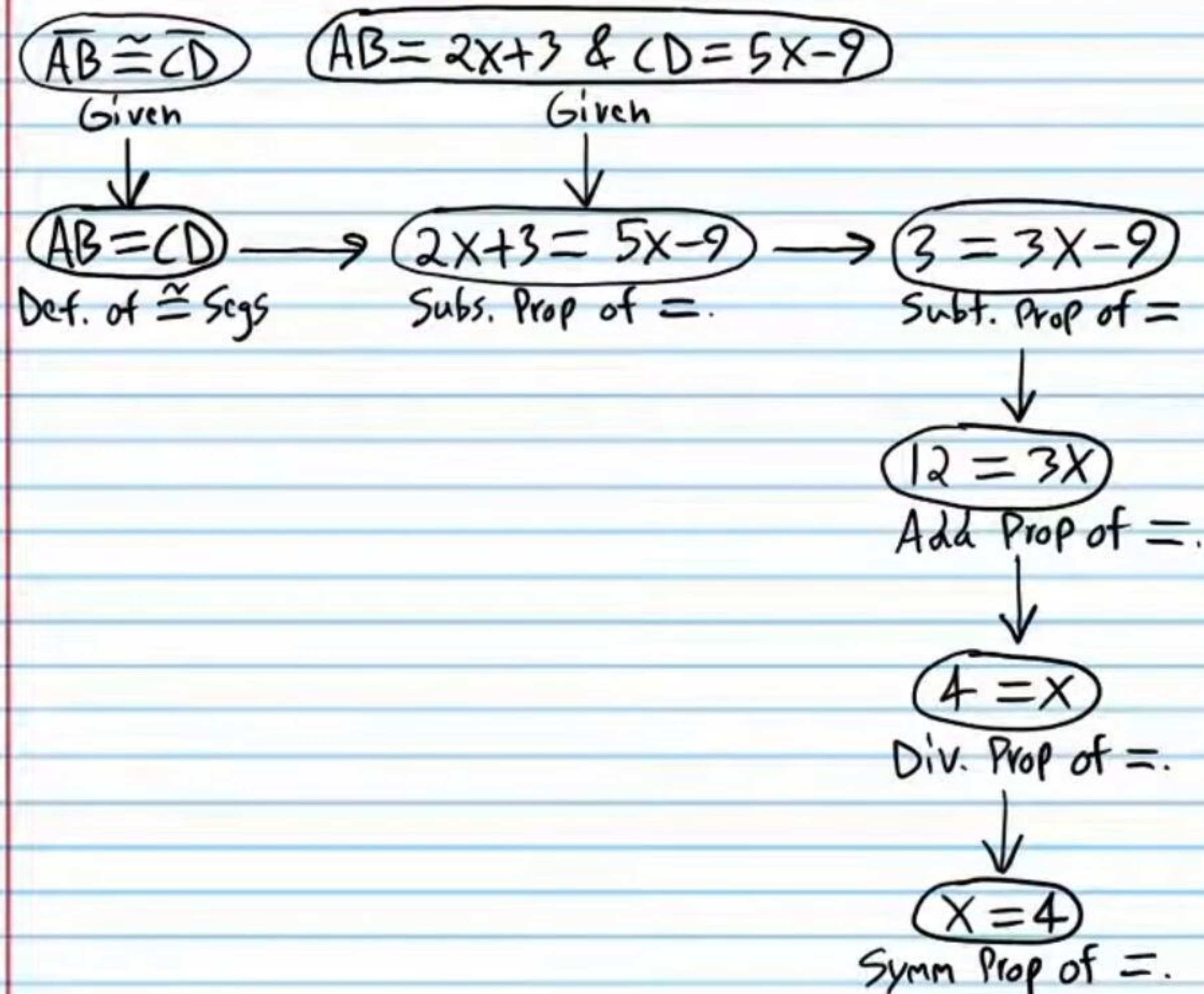
⑦ If the two segments below are congruent, then $x=4$.



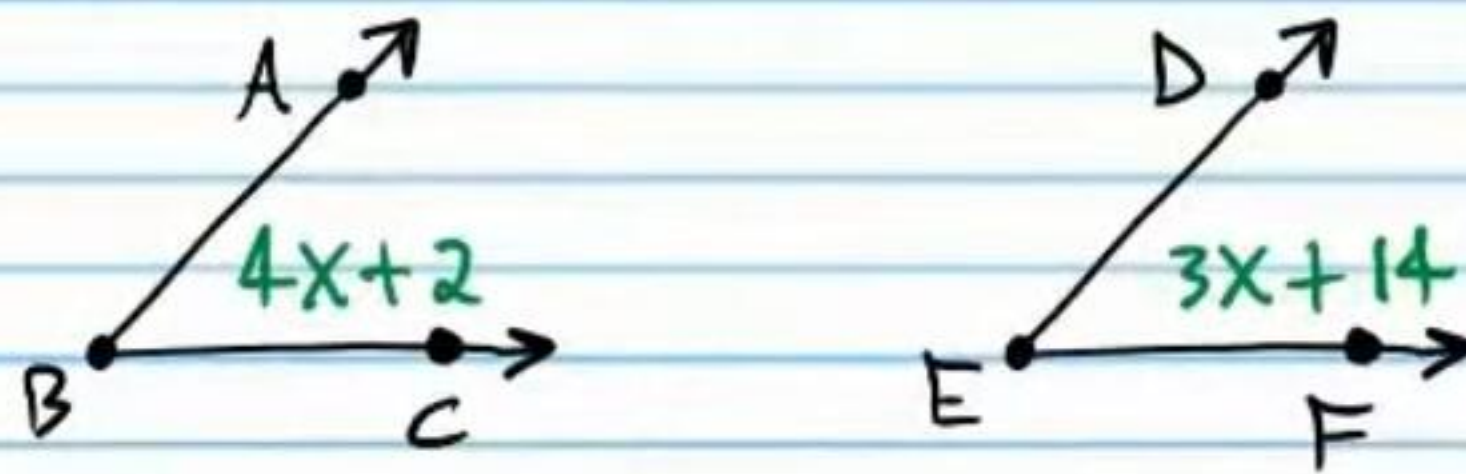
Two-Column Form

Statement	Reason
I) $\overline{AB} \cong \overline{CD}$	Given.
II) $AB = CD$	Def. of $\cong$ Segs.
III) $AB = 2x+3$ & $CD = 5x-9$.	Diagram.
IV) $2x+3 = 5x-9$	Subs. Prop of =.
V) $3 = 3x-9$	Subt. Prop of =.
VI) $12 = 3x$	Add. Prop of =.
VII) $4 = x$	Div. Prop of =.
VIII) $x = 4$	Symm. Prop of =.

Flowchart Form



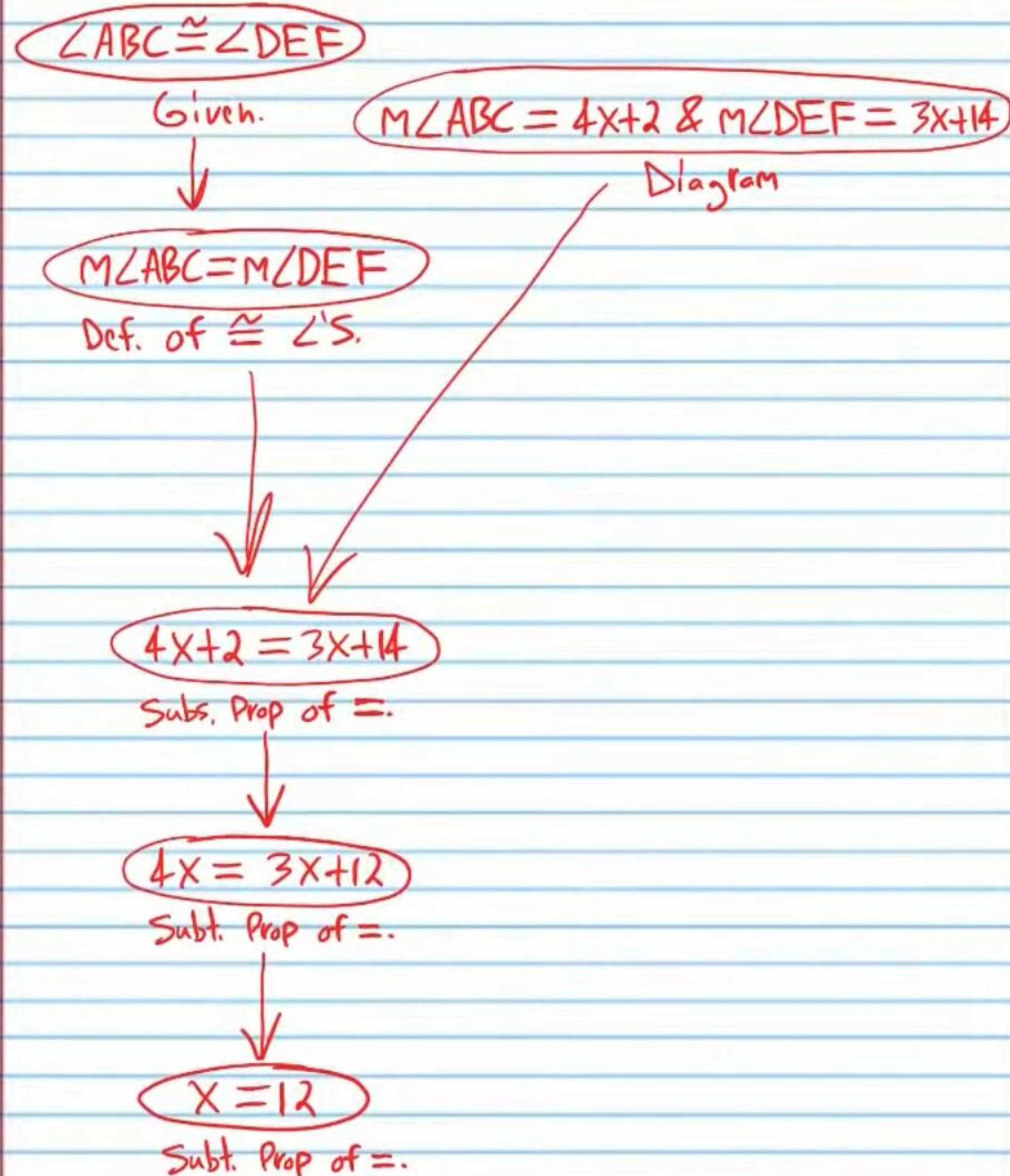
⑧ If the two angles below are congruent, then $x = 12$.



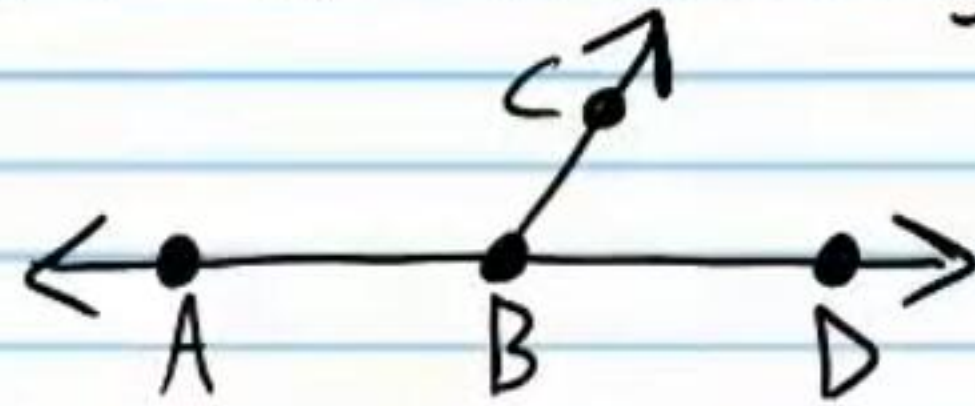
Two-Column Form

Statement	Reason
I) $\angle ABC \cong \angle DEF$	Given.
II) $m\angle ABC = m\angle DEF$	Def. of $\cong$ $\angle$'s.
III) $m\angle ABC = 4x+2$ and $m\angle DEF = 3x+14$.	Given (diagram).
IV) $4x+2 = 3x+14$	Subs. Prop of $=$.
V) $x+2 = 14$	Subt. Prop of $=$.
VI) $x = 12$	Subt. Prop of $=$.

Flowchart Form



⑨ If $m\angle ABC = 130^\circ$ in the diagram below, then $m\angle CBD = 50^\circ$.



Two-Column Form

Statement	Reason
I) $\angle ABC$ & $\angle CBD$ form a linear pair.	Diagram.
II) $\angle ABC$ & $\angle CBD$ are supplementary.	Linear Pair Post.
III) $m\angle ABC + m\angle CBD = 180^\circ$	Def. of supp $\angle$'s.
IV) $m\angle ABC = 130^\circ$	Given.
V) $130 + m\angle CBD = 180$	Subs. Prop of =.
VI) $m\angle CBD = 50^\circ$	Subt. Prop of =.

Flowchart Form

$\angle ABC$ & $\angle CBD$ form a linear pair.

Diagram

$\angle ABC$ & $\angle CBD$ are supp. $\angle$'s

Linear Pair Post

$m\angle ABC + m\angle CBD = 180^\circ$

Def. of supp. $\angle$'s.

$130 + m\angle CBD = 180$

Subs. Prop of =.

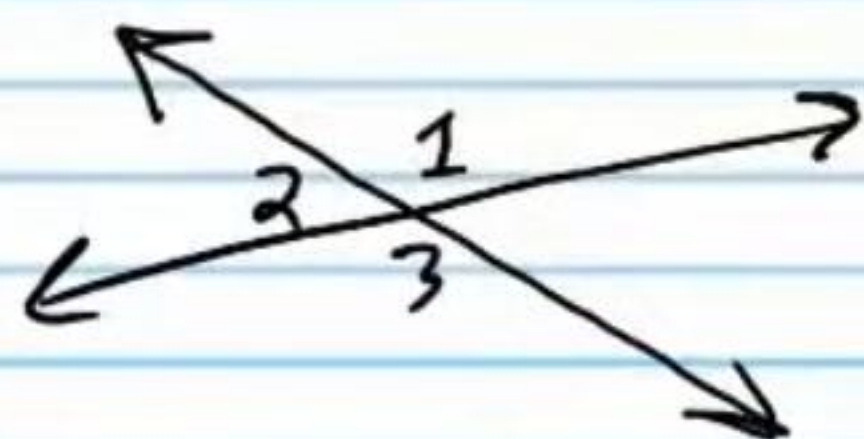
$m\angle CBD = 50^\circ$

Subt. Prop of =.

$m\angle ABC = 130^\circ$

Given

⑩ Vertical angles are congruent (i.e. $\angle 1 \cong \angle 3$).



Two-Column Form

Statement	Reason
I) $\angle 1$ & $\angle 3$ are Vert. $\angle$'s	Diagram
II) $\angle 2$ & $\angle 1$ form a linear pair and $\angle 2$ & $\angle 3$ form a linear pair.	Diagram
III) $\angle 2$ & $\angle 1$ are supplementary and $\angle 2$ & $\angle 3$ are supplementary.	Linear Pair Post.
IV) $\angle 1 \cong \angle 3$	$\cong$ Supps. Thm.

Flowchart Form

$\angle 1$ & $\angle 3$ are Vert. $\angle$'s

Diagram

$\angle 2$ & $\angle 1$ form a linear pair and $\angle 2$ & $\angle 3$ form a linear pair.

Diagram

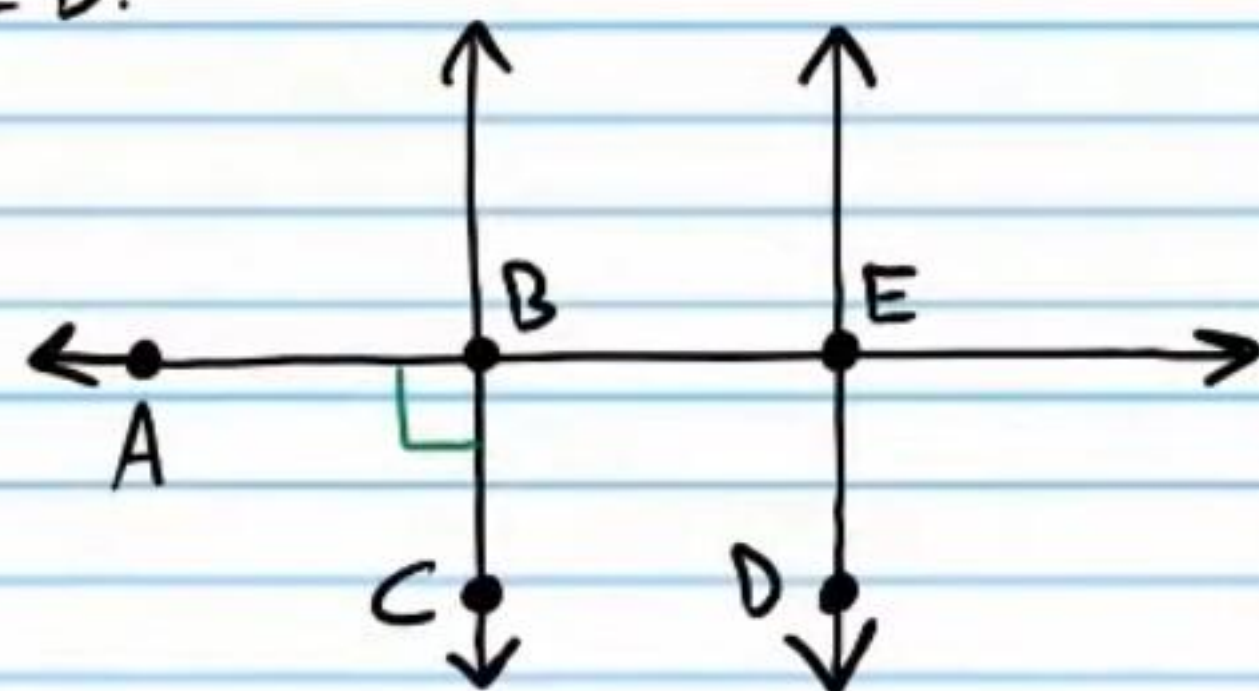
$\angle 2$ & $\angle 1$ are supplementary and $\angle 2$ & $\angle 3$ are supplementary.

Linear Pair Postulate

$\angle 1 \cong \angle 3$

$\cong$ Supps. Thm.

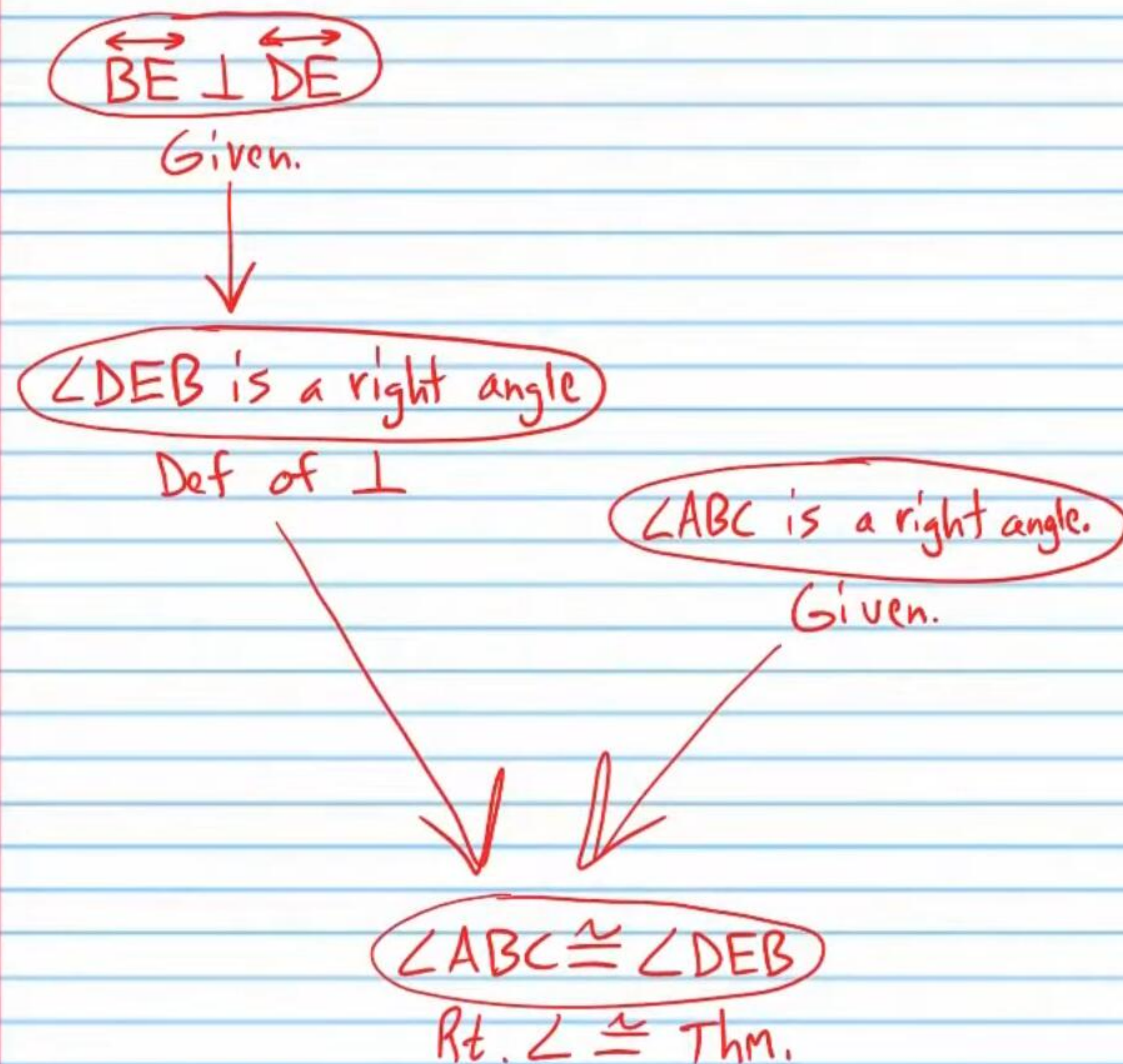
⑪ If $\overleftrightarrow{BE} \perp \overleftrightarrow{DE}$ and $\angle ABC$ is a right angle,
then $\angle ABC \cong \angle DEB$.



Two-Column Form

Statement	Reason
I) $\overleftrightarrow{BE} \perp \overleftrightarrow{DE}$	Given.
II) $\angle DEB$ is a right angle.	Def. of $\perp$
III) $\angle ABC$ is a right angle.	Given.
IV) $\angle ABC \cong \angle DEB$	Rt. $\angle \cong$ Thm.

Flowchart Form



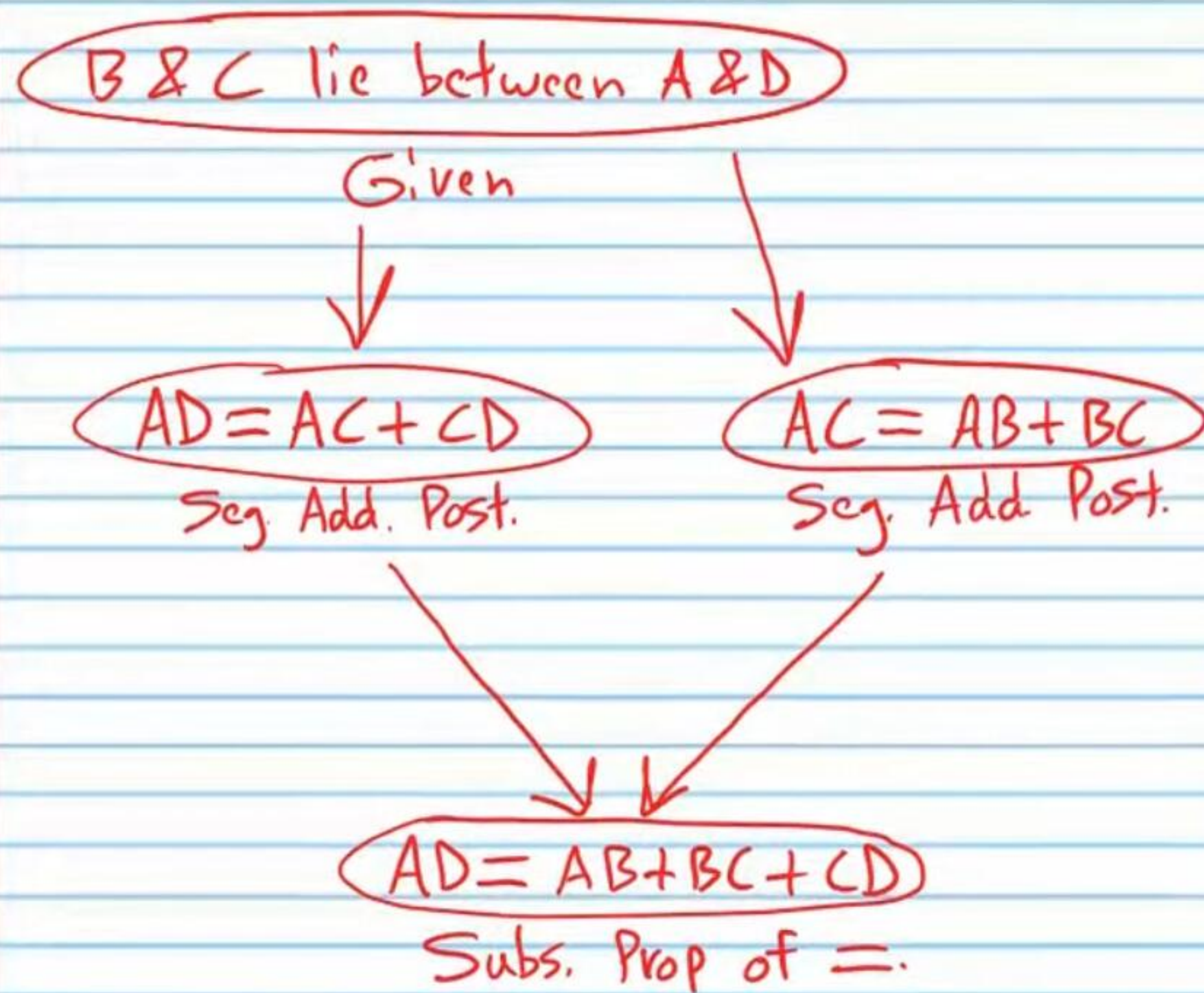
(12) IF B and C lie between A and D (collinear) on the segment below, then $AD = AB + BC + CD$.



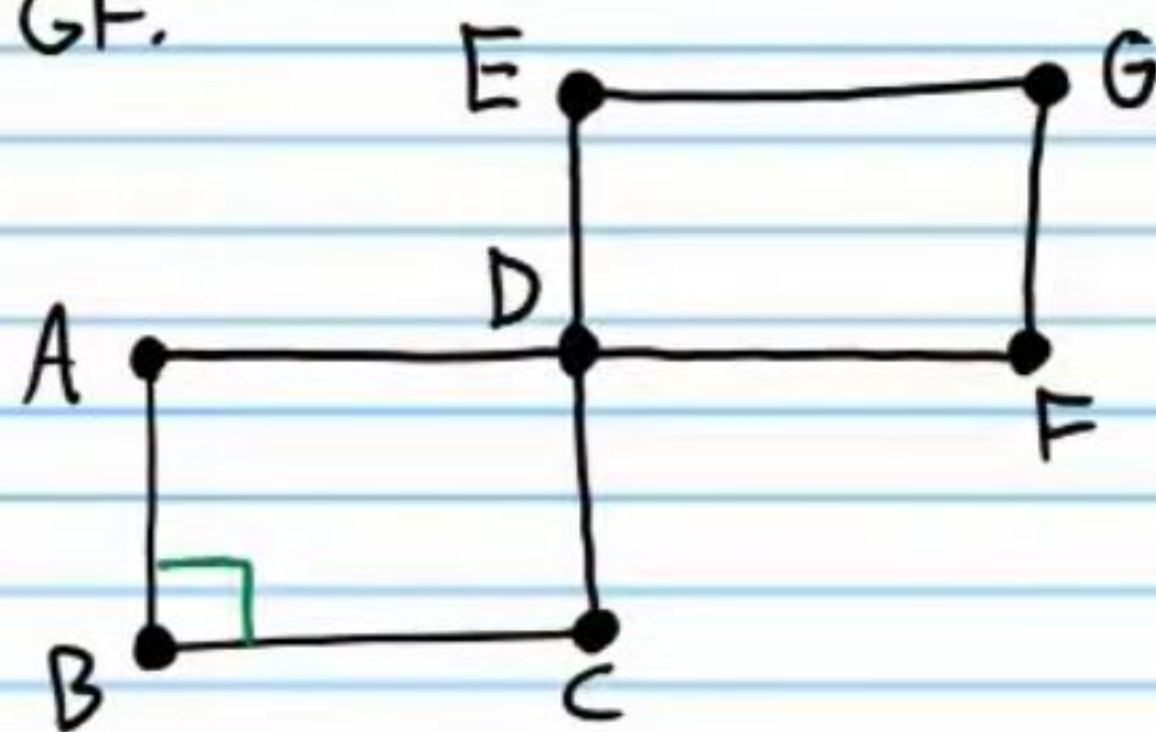
Two-Column Form

Statement	Reason
I) B & C lie between A & D.	Given.
II) $AD = AC + CD$	Seg. Add. Post
III) $AC = AB + BC$	Seg. Add. Post
IV) $AD = AB + BC + CD$	Subs. Prop of =.

Flowchart Form



(13) IF $m\angle ABC = m\angle ADC$ below and $m\angle EDF = m\angle EGF$, then $\overline{EG} \perp \overline{GF}$.



Two-Column Form

Statement	Reason
I) $\angle ABC$ is a right angle.	Diagram.
II) $m\angle ABC = 90^\circ$	Def of rt. $\angle$.
III) $m\angle ABC = m\angle ADC$	Given.
IV) $m\angle ADC = 90^\circ$	Trans. Prop of =.
V) $m\angle ADC = m\angle EDF$	Vert. $\angle$'s Thm.
VI) $m\angle EDF = 90^\circ$	Trans. Prop of =.
VII) $m\angle EDF = m\angle EGF$	Given.
VIII) $m\angle EGF = 90^\circ$	Trans. Prop of =.
IX) $\angle EGF$ is a right angle.	Def. of rt. $\angle$.
X) $\overline{EG} \perp \overline{GF}$	Def of $\perp$.

Flowchart Form

$\angle ABC$ is a right angle

Diagram

$m\angle ABC = 90^\circ$

Def. of rt $\angle$.

$m\angle ABC = m\angle ADC$

Given.

$m\angle ADC = 90^\circ$

Trans. Prop of =.

$m\angle ADC = m\angle EDF$

Vert. $\angle$'s Thm

$m\angle EDF = 90^\circ$

Trans. Prop of =

$m\angle EDF = m\angle EGF$

Given

$m\angle EGF = 90^\circ$

Trans. Prop of =.

$\angle EGF$ is a right angle

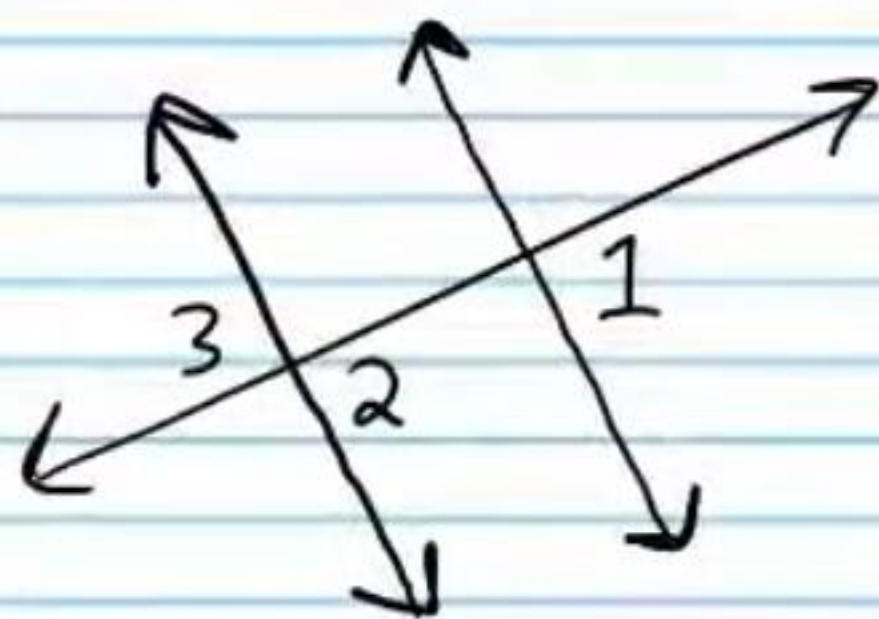
Def of rt. $\angle$.

$\overline{EG} \perp \overline{GF}$

Def of $\perp$.

Write flowchart proofs for the following statements.

- ⑭ If $m\angle 1 = m\angle 2$ in the figure, then $m\angle 1 = m\angle 3$.
 ★ You will need the Vertical Angles Theorem.

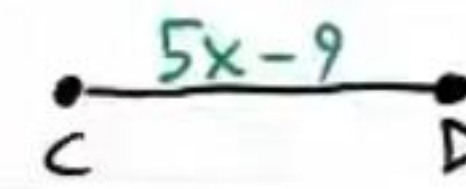
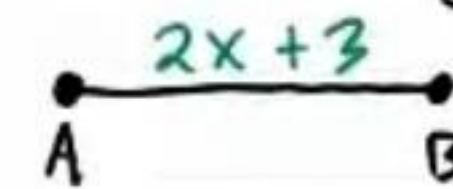


$m\angle 1 = m\angle 2$
 Given

$m\angle 3 = m\angle 2$
 Vert. $\angle$'s Thm

$m\angle 1 = m\angle 3$
 Trans. Prop of =.

- ⑦ If the two segments below are congruent, then $x=4$.



- ⑮ The converse of problem #7.

$x=4$
 Given.

$AB = 2x+3$ & $CD = 5x-9$
 Diagram.

$AB = 2(4)+3$ & $CD = 5(4)-9$
 Subs. Prop of =.

$AB = 11$ & $CD = 11$
 Simplification.

$AB = CD$
 Trans. Prop of =.

$\overline{AB} \cong \overline{CD}$
 Def of $\cong$ Segs.

(16) If point B bisects $\overline{AC}$, then $AC=10$.



B bisects $\overline{AC}$.

Given.

$AB = x+2$ & $BC = 2x-1$

Diagram

$AB = BC$

Def of seg bisector.

$x+2 = 2x-1$

Subs. Prop of =.

$2 = x-1$

Subt. Prop of =.

$3 = x$

Add Prop of =.

$AC = 3(3)+1$

Subs. Prop of =

$AC = 10$

Simplification.

$AC = AB + BC$

Seg. Add. Post.

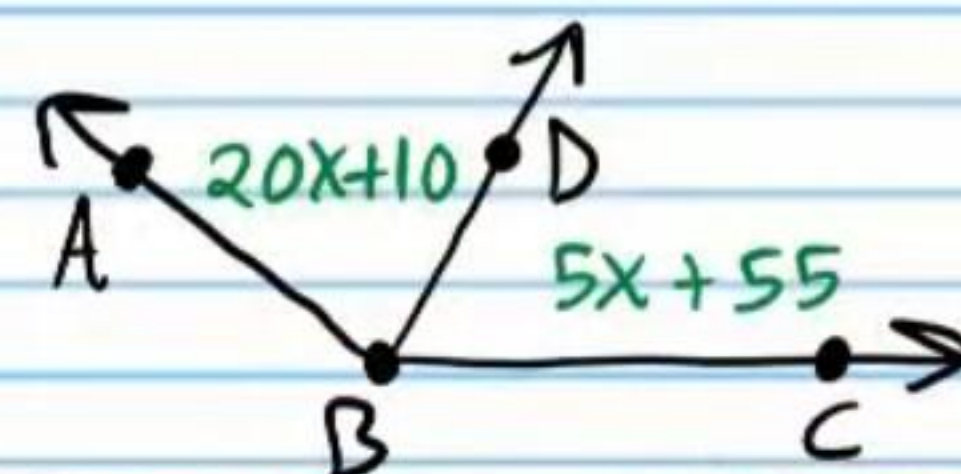
$AC = x+2 + 2x-1$

Subs. Prop of =.

$AC = 3x+1$

Simplification.

(17) If $m\angle DBC = 70^\circ$, then $\overrightarrow{BD}$ bisects $\angle ABC$.



$m\angle DBC = 70^\circ$

Given.

$m\angle DBC = 5x+55$

Diagram.

$5x+55 = 70$

Trans. Prop of =.

$5x = 15$

Subt. Prop of =.

$x = 3$

Div. Prop of =.

$m\angle ABD = 20x+10$

Diagram

$m\angle DBC = 5(3)+55$ &
 $m\angle ABD = 20(3)+10$

Subs. Prop of =.

$m\angle DBC = 70^\circ$ & $m\angle ABD = 70^\circ$

Simplification.

$m\angle DBC = m\angle ABD$

Trans. Prop of =.

$\overrightarrow{BD}$ bisects $\angle ABC$

Def of $\angle$ bisector.